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CS 210 Project Three – Corner Grocer

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The Grocery Tracker program is designed to help users track the frequency of grocery items in their shopping lists. This document provides an overview of the design and functionality of the program.

Program Structure Overview:

The cornerstone of this program is the **GroceryTracker** class, which orchestrates the essential operations for managing item frequencies and coordinating user interactions. The main function acts as the gateway to the program, orchestrating user interactions with the menu options and facilitating communication with the **GroceryTracker** instance. Data ingestion occurs through the parsing of an input file named **"CS210_Project_Three_Input_File.txt,"** populating a map named **itemFrequency** with the relevant data. Users are presented with a menu, offering a suite of options including item search, frequency printing, histogram display, and program exit. These user selections trigger corresponding methods within the **GroceryTracker** class to execute the desired functionalities. A vital feature of the program is its capability to safeguard data integrity by facilitating backup operations to a file named **"frequency.dat."**

Implementation Nuances:

Within the GroceryTracker class, the discrete **itemFrequency** map serves as the repository for item names and their respective occurrence frequencies. The **readDataFromFile** method orchestrates the parsing of the input file, iteratively updating the itemFrequency map with the encountered items and their frequencies. For user-requested item inquiries, the **printItemFrequency** method efficiently retrieves and presents the frequency of the specified item from the **itemFrequency** map. The **printAllItemFrequency** method systematically traverses the itemFrequency map, furnishing a comprehensive listing of all items alongside their respective frequencies.

Utilizing the **printItemHistogram** method, the program renders a visual representation of item frequencies in the form of a histogram, aiding in intuitive data interpretation.

To preserve data integrity, the **backupData** method secures a snapshot of the current item frequencies in the designated "frequency.dat" file, ensuring data persistence beyond program termination.

Menu Interaction Dynamics:

Employing a structured menu interface, the main function orchestrates user interactions via a robust do-while loop mechanism, ensuring continued engagement until program exit.

User inputs, solicited within the menu interface, are meticulously processed, with the program guiding users through the available options and executing their selections accordingly.

Menu selections are mapped to corresponding actions through a switch-case construct, orchestrating seamless navigation between functionalities. Upon user request for item search, frequency printing, histogram display, or program exit, the program delegates execution to the relevant methods within the **GroceryTracker** class.

In essence, the provided C++ program for the Corner Grocer item-tracking system embodies a meticulously structured solution, adeptly managing item frequencies and facilitating user interactions through a menu-driven interface. However, to bolster its robustness and user-friendliness, the implementation would benefit from incorporating comprehensive error handling and input validation mechanisms.